Solid State Physics — Week 6 Assignment Questions & Answers

Question 51

If the x-rays of wavelength 0.5 Å are diffracted at an angle of 5° in the first order, what is the spacing between the adjacent planes of the crystal?

(a) 1.26 Å

(b) 2.87 Å

(c) 3.12 Å

(d) 4.27 Å

Accepted answer: (b)

Question 52

The first order (100) reflection angle is 18° for a simple cubic crystal using x-rays of wavelength 1.54Å. Determine the distance between (111) planes of the crystal.

(a) 1.44 Å

(b) 2.82 Å

(c) 3.12 Å

(d) 4.23 Å

Accepted answer: (a)

Question 53

The x-ray of wavelength 0.563 Å from a silver target is analyzed with a Bragg's spectrometer using a calcite crystal. Consider calcite is a simple cubic crystal having lattice parameter 3.03Å. What is the highest order for which the diffraction line may be observed?

(a) 4

(b) 6

(c) 10

(d) 14

Accepted answer: (c)

Question 54

In a powder diffraction experiment with x-rays of wavelength λ = 1.54 Å, the first order Bragg reflection occur from a monatomic cubic crystal for θ values: 17.4°, 20.3°, 29.3°, 35.1° and 36.9°. Determine the ratio of Miller indices of diffraction planes.

(a) (111), (200), (220), (311), (222)

(b) (100), (210), (221), (322), (232)

(c) (110), (111), (211), (223), (332)

(d) (121), (112), (252), (313), (333)

Accepted answer: (a)

Question 55

Kcl crystallizes as such it is viewed as simple cubic structure and within the unit cell contain one formula unit of Kcl. Find the glancing angle at which an x-ray of wavelength λ = 1.787Å is reflected from (100) planes in the third order. Given: density of Kcl = 1990 kg/m^3 and its molecular weight is 74.6 gm/mol.

(a) 22.6°

(b) 42.6°

(c) 64.5°

(d) 74.8°

Accepted answer: (b)

Question 56

A powder camera of radius 57.3 mm is used to obtain diffraction pattern of gold (FCC) having a lattice parameter of 4.08Å. The wavelength of x-ray λ = 0.71Å is used. Calculate the first two S-values, where S is the distance between the arcs on the x-ray photographic plate. Consider the first order Bragg diffraction.

(a) $34.68\ mm, 40.08\ mm$

(b) $44.24\ mm, 51.22\ mm$

(c) $52.53\ mm, 62.42\ mm$

(d) $64.28\ mm, 74.78\ mm$

Accepted answer: (a)

Question 57

For the first order Bragg diffraction, the S values for the first three lines in the powder pattern of a cubic crystal are 56.8, 94.4 and 112.0 mm respectively. Given the wavelength of x-radiation used is 0.71Å and the camera radius 57.30 mm, determine the lattice parameter of the crystal.

(a) 2.51 Å

(b) 2.87 Å

(c) 3.42 Å

(d) 3.85 Å

Accepted answer: (a)

Question 58

The primitive translation vectors of a two dimensional lattice are $\vec{a} = 2\hat{\imath} + \hat{\jmath}, \vec{b} = 2\hat{\jmath}$. Determine the primitive translational vectors of its reciprocal lattice.

(a) $\vec{a}^* = \pi\hat{\imath}, \vec{b}^* = \frac{\pi}{2}(2\hat{\jmath} - \hat{\imath})$

(b) $\vec{a}^* = 2\pi\hat{\imath}, \vec{b}^* = \pi(2\hat{\jmath} - \hat{\imath})$

(c) $\vec{a}^* = 3\pi\hat{\imath}, \vec{b}^* = \frac{\pi}{2}(2\hat{\jmath} + \hat{\imath})$

(d) $\vec{a}^* = 4\pi\hat{\imath}, \vec{b}^* = \pi(\hat{\imath} - 2\hat{\jmath})$

Accepted answer: (a)

Question 59

The primitive basis vectors of a three dimensional lattice are $\vec{a} = \hat{\imath} + 2\hat{\jmath}$; $\vec{b} = 4\hat{\jmath}$ and $\vec{c} = \hat{k}$. Find out the primitive translational vectors of its reciprocal lattice.

(a) $\vec{a}^* = 2\pi\hat{\imath}, \vec{b}^* = \frac{\pi}{2}(\hat{\jmath} - 2\hat{\imath}), \vec{c}^* = 2\pi\hat{k}$

(b) $\vec{a}^* = \pi\hat{\imath}, \vec{b}^* = \pi(\hat{\jmath} - 2\hat{\imath}), \vec{c}^* = \pi\hat{k}$

(c) $\vec{a}^* = 3\pi\hat{\imath}, \vec{b}^* = \frac{\pi}{3}(\hat{\jmath} - 2\hat{\imath}), \vec{c}^* = 3\pi\hat{k}$

(d) $\vec{a}^* = 4\pi\hat{\imath}, \vec{b}^* = \frac{\pi}{4}(\hat{\jmath} - 2\hat{\imath}), \vec{c}^* = 4\pi\hat{k}$

Accepted answer: (a)

Question 60

A copper wire of 1 mm diameter carries a current of 5 mA. Calculate the average drift velocity of the free electrons. Copper is monovalent and crystallizes in FCC structure with a lattice parameter of 3.61Å.

(a) $2.38 \times 10^{-7}\ m/s$

(b) $4.68 \times 10^{-7}\ m/s$

(c) $6.28 \times 10^{-7}\ m/s$

(d) $8.39 \times 10^{-7}\ m/s$

Accepted answer: (b)